Pasin Makcharoen - 6810545794

Overview

main.py is a command-line student management program using MongoDB. A menu lets the user add, view, search (by ID, GPA, or department with $and / $or), update, and delete students.

Static Analysis Results

Mypy checks that values match their type annotations without running the code. It skips functions that have no annotations. Pylint checks coding standards, likely mistakes and code quality, and gives a score out of 10.

Tool	Finding	Why it matters
Mypy	“Success: no issues”	Misleading: nothing was annotated, so function bodies were skipped. With --check-untyped-defs it finds 2 errors.
Pylint	W0718 broad-exception-caught (5)	except Exception also hides programming mistakes.
Pylint	W0612 unused variables, W0611 unused import	Dead code.
Pylint	C0303 trailing whitespace (23), W0511 TODO (16), C0301 long lines (2)	Style and readability.

Code Improvements

Fix 1: catch a specific error (Pylint W0718) and add type annotations so Mypy checks the function.

Fix 2: a bug neither tool reported. A MongoDB cursor is always truthy, so “No students found.” never printed.

Also removed unused code, trailing whitespace and TODO comments, sorted imports, and wrapped long lines.

Status	Mypy	Pylint
Before	0 issues (2 errors with --check-untyped-defs)	6.45 / 10
After	0 issues	9.24 / 10

Comparison and Reflection

Mypy checks types, and pylint checks style and common mistakes. They find different problems, so they complement each other, Pylint found the broad exceptions and unused code, and Mypy found a variable reused with two types. Both have limits. Mypy only checks annotated code, and neither tool found the empty-cursor logic bug, which needs testing. Static analysis catches cheap mistakes early and keeps code consistent.